LectureLens

AI-Powered Lecture Capture System

Complete Documentation & User Guide

Version 3.0

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Powered by Anthropic Claude & Google Gemini

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1. Overview

LectureLens is a plug-and-play AI lecture capture system that runs entirely from a USB drive. It records classroom whiteboards and projector screens using the document camera already installed in every classroom, automatically generates detailed AI notes, answers instructor questions in real time, and produces three professionally formatted HTML files at the end of every session — all without requiring any software installation on the school computer.

1.1 What it does

- Captures photos of the whiteboard at the press of a button or presenter remote
- Takes screenshots of slides, websites, or any screen content with the S key
- Sends each capture through a two-pass AI pipeline that first reads the board, then writes the notes
- Smart Lookback compares captures and removes blurry or duplicate shots automatically
- Lets the teacher type questions (N key) or photo-questions (O key) and receive AI answers
- Saves all notes, photos, and Q&A to a dated session folder on the USB
- Generates three HTML files with full LaTeX math rendering via MathJax when the session ends
- Recovers gracefully if the laptop closes early using backup text files

1.2 What's new in Version 3.0

> Version 3.0 is a major accuracy and cost release. The headline change is a **hybrid AI engine**: ordinary camera captures are now read by Google Gemini Flash-Lite, while Anthropic Claude Sonnet still handles the work that benefits most from its reasoning. The result is roughly **60% lower cost per lecture** with equal or better reading accuracy.

- **Hybrid AI engine** NEW — camera captures use Gemini Flash-Lite; Claude Sonnet handles Smart Lookback, screenshots, notes, and the final summary.
- **Lossless PNG capture** NEW — photos are saved as PNG instead of compressed JPEG, eliminating the compression artifacts that previously caused misread signs and digits in equations.
- **Smart brightness correction** IMPROVED — CLAHE adaptive brightness now runs only when the room is genuinely dark, so it no longer washes out faint marker strokes on a well-lit board.
- **Content-aware cropping** IMPROVED — the system now detects where the board and writing actually are and sends focused crops, instead of fixed overlapping crops.
- **Gated Smart Lookback** IMPROVED — Smart Lookback now runs only when there is a real reason (a retake or an unclear note), instead of after every single capture, cutting cost significantly.
- **Screenshot verification pass** NEW — screenshot notes get a final audit pass that checks the math by substitution before saving.
- **Rotation controls** NEW — the R and T keys rotate future captures, useful when the camera or a document is not aligned.
- **Hardened recovery tool** IMPROVED — Recover.exe now correctly rebuilds Smart-Lookback-cleaned notes and handles all backup file formats robustly.

1.3 Who it is for

LectureLens is designed for teachers and educators who want to provide students with accurate, detailed lecture notes without the effort of manual note-taking. It is especially useful for subjects involving equations, diagrams, and complex whiteboard content such as mathematics, physics, chemistry, and engineering.

2. Hardware Requirements

2.1 What you need

Item	Purpose	Estimated Cost
Classroom document camera	Captures the whiteboard and worksheets	Already installed in every room
Wireless presenter remote (clicker)	Teacher triggers captures wirelessly	~$20
USB flash drive (32GB+)	Runs the program and stores all data	~$10
School computer with internet	Runs the exe and sends API calls	Already available

> LectureLens uses the document camera that is already installed in every classroom, so there is no webcam to buy, no USB extension cable to run, and no ceiling bracket to mount. The only items the teacher needs are a presenter remote and the LectureLens USB drive — both of which travel with them between rooms.

2.2 Camera notes

Any standard classroom document camera that connects to the computer over USB will work — LectureLens detects it automatically. Aim the camera at the whiteboard, or place a worksheet or textbook page under it. Version 3.0 includes smart CLAHE brightness correction that automatically adjusts for dark or unevenly lit classrooms — and, unlike earlier versions, it leaves well-lit boards untouched so faint marker strokes are preserved. If the camera image comes through rotated, the R and T keys correct the orientation (see section 5.5).

2.3 Physical setup

1. Confirm the classroom document camera is connected to the computer and powered on
2. Aim the document camera at the whiteboard, or position it over the desk for worksheets
3. Plug the presenter remote USB dongle into a USB port
4. Plug the LectureLens USB drive into another USB port
5. Double-click `LectureLens.exe` on the USB drive to start

The document camera stays in the classroom. Only the USB drive and presenter remote travel with the teacher between rooms.

3. Software & Files

3.1 What is on the USB drive

File	Description
`LectureLens.exe`	Main program — double-click to start a lecture session
`Recover.exe`	Recovery tool — rebuilds HTML files from backup notes if the laptop closed early
`config.txt`	Holds the API keys for Claude and Gemini (see 3.2)
`sessions/`	Folder where all session data is automatically saved

3.2 API keys & config.txt

LectureLens needs an Anthropic API key, and optionally a Google Gemini key for the hybrid engine. The simplest method is a plain text file named `config.txt` on the USB drive:

- `ANTHROPIC_API_KEY=your_key_here` — required for Smart Lookback, screenshots, notes, and the summary
- `GEMINI_API_KEY=your_key_here` — optional; enables the low-cost hybrid engine for camera captures

> If no Gemini key is found, LectureLens automatically falls back to using Claude Sonnet for camera captures. Everything still works — it simply costs more per lecture. The program prints a one-line notice at startup when this happens.

3.3 Session folder structure

Every lecture creates a new dated and named folder inside `sessions/`. Claude names it based on the lecture content after the session ends.

Path	Contents
`sessions/2026-05-16_09-30_linear-algebra/`	Root session folder (renamed by Claude after the session)
`01_photos.html`	All captured whiteboard photos only (no screenshots)
`02_ai_notes.html`	Claude's notes for every capture + screenshots + Q&A in chronological order
`03_summary.html`	Full lecture summary with LaTeX math rendered by MathJax
`images/`	Raw PNG files for every photo and screenshot captured
`backups/`	Text files saved after each AI response (crash recovery)
`backups/replaced_captures/`	Captures removed by Smart Lookback (kept as backup)
`backups/duplicate_skipped_captures/`	Near-duplicate photos skipped by the local pre-filter before any AI call

> In Version 3.0 raw captures are saved as **lossless PNG** rather than JPEG. PNG files are larger, but they preserve the exact pixels of thin equation strokes — this is the single biggest reason v3.0 misreads fewer signs and digits.

3.4 The three HTML files explained

01_photos.html

Contains every raw whiteboard photo captured during the lecture, one per section, with a timestamp header. Screenshots taken with the S key are NOT included here — they appear in `02_ai_notes.html` only. Open in any browser to review exactly what was on the board at each point.

02_ai_notes.html

Contains the AI's detailed notes for every capture in chronological order — exactly as the lecture progressed. Camera captures (blue header), screenshots (green header), text notes from the N key (orange header), and photo notes from the O key (orange header) all appear in the order they were taken. All math is rendered as proper typeset equations via MathJax. Open in any browser, or print to PDF using Ctrl+P.

03_summary.html

Contains a comprehensive lecture summary generated by Claude after all captures are processed. Includes the main topic, learning objectives, key equations, step-by-step procedures, major takeaways, and a Questions and Answers from Class section. All math rendered with MathJax.

4. The Hybrid AI Engine

4.1 Why two models

Version 3.0 splits the AI workload between two providers to balance cost and quality. Each model is used where it performs best:

Task	Model	Why
Ordinary camera captures	Google Gemini Flash-Lite	High-volume, routine board reading — far cheaper per image at comparable accuracy
Smart Lookback comparisons	Claude Sonnet	Needs careful multi-image reasoning to decide what to keep or hide
Screenshots (slides, websites)	Claude Sonnet	Mixed content types and a verification pass benefit from stronger reasoning
Text & photo notes (N / O keys)	Claude Sonnet	Open-ended teacher questions
Final lecture summary	Claude Sonnet	Synthesizes the whole session into structured notes
Session & folder naming	Claude Haiku	Tiny task; the fastest, cheapest model is plenty

4.2 Two-pass reading

Every capture is processed in two passes rather than one. **Pass 1 (Read)** transcribes exactly what is on the board — every coefficient, sign, and matrix entry — without solving anything. **Pass 2 (Solve)** takes that transcription and produces clean, structured notes, verifying any final answer by substitution. Separating reading from solving keeps the model from "fixing" the board to match what it expects.

4.3 Image accuracy pipeline

Before any capture reaches the AI, Version 3.0 prepares it for the most accurate possible read:

- **Lossless PNG** — no JPEG compression artifacts around thin strokes.
- **Conditional brightness correction** — CLAHE runs only on genuinely dark frames; bright boards pass through untouched.
- **Content-aware crops** — the system detects the bright board region and the dense-writing region, and sends focused, zoomed crops of exactly those areas alongside the full frame.
- **Resolution matching** — images are sized to the model's effective resolution, so no detail is lost and no tokens are wasted.

> Most of these steps are fully automatic and need no teacher action. They simply make the notes in `02_ai_notes.html` more accurate — especially for dense systems of equations and matrices.

5. Controls & Keyboard Shortcuts

5.1 Key reference

Key / Button	Action	Details
SPACE	Capture photo	Takes a photo instantly, queues it for AI processing
Remote clicker (right arrow)	Capture photo	Same as SPACE — works from anywhere in the room
iClicker button (F5)	Capture photo	Multiple iClickers can be used simultaneously
S	Screenshot + note	Captures the full screen, optional note, AI analyzes content type and writes notes
N	Text note	Type a question; Claude answers in the terminal and HTML
O	Photo note	Snaps a board photo, type a question; Claude answers with board context
R	Rotate 180°	Toggles 180° rotation for future captures; press again to reset
T	Rotate 90°	Rotates future captures +90° clockwise each press (0→90→180→270)
TAB	Switch camera	Toggles cameras if a second USB camera is connected (normally just the doc cam)
ESC	Stop session	Waits for the queue to finish, then generates all 3 HTML files

5.2 Capture workflow (SPACE / remote)

1. Photo taken instantly and saved to the `images/` folder as lossless PNG
2. Brightness correction is applied automatically if the room is dark
3. A local pre-filter checks whether the photo is a near-duplicate of the last one
4. Green flash border appears on the live preview window
5. Photo added to the processing queue
6. The hybrid engine reads and writes notes in the background (typically 3–8 seconds)
7. Smart Lookback runs if a retake or unclear note triggers it
8. Notes saved to the `backups/` folder as a text file
9. Live preview shows updated counts: `Done: X Queue: Y`

5.3 Screenshot workflow (S key)

1. Full screen captured silently and saved to `images/` as PNG
2. Preview minimizes automatically
3. Terminal asks whether to add a note about the screenshot (Enter to skip; auto-cancels in 15s)
4. Claude identifies the content type (slide, website, code, document, etc.)
5. Claude writes structured notes based on the content type
6. A verification pass checks any math by substitution before saving NEW
7. Preview restores; the screenshot and notes appear in `02_ai_notes.html` with a green header

5.4 Note & photo-note workflow (N / O keys)

N key (text note): the preview minimizes, the terminal asks for your question, and Claude generates a written answer in a few seconds. Spaces alone do not reset the 15-second auto-cancel timer — only real characters count. The note and answer are saved to `backups/` and included in `02_ai_notes.html`.

O key (photo note): a board photo is taken silently first, then you type your question. Claude answers using both the question and the board photo for full context. The answer appears in the terminal with a board-context label.

5.5 Rotation controls (R / T keys)

> New in Version 3.0. If the camera is mounted at an angle, or you are capturing a textbook page that is sideways, press **R** to flip future captures 180°, or **T** to rotate them 90° clockwise (press repeatedly to cycle). Rotation applies to all subsequent captures and screenshots until changed. Press R again, or cycle T back to 0°, to return to normal.

5.6 Smart Lookback & duplicate pre-filter

Version 3.0 has two layers that keep your notes clean and your costs down:

Local duplicate pre-filter (free)

Before any AI call, LectureLens compares each new photo against the previous one using a fast local image check. If the photo is a near-identical retake, it is skipped entirely — no API cost — and moved to `backups/duplicate_skipped_captures/`. A retake that is genuinely sharper is kept and passed on.

Smart Lookback (Claude Sonnet)

When a capture is kept, Smart Lookback can compare it against the last few captures and hide an older one that is blurry, unclear, or fully replaced by the newer shot. The hidden capture is moved to `backups/replaced_captures/` and is never permanently deleted.

- In Version 3.0, Smart Lookback runs only when there is a real reason — a sharper retake, or notes that look unclear — instead of after every capture.
- It can also use mathematical reasoning to fill in `[unclear]` values where the rest of the work makes them certain.
- Disable it with `SMART_LOOKBACK_REPAIR = False`; disable only the hiding behaviour with `SMART_DELETE_WORSE_CAPTURES = False`.

6. Cost Breakdown

6.1 Hardware — one-time cost

Because LectureLens uses the document camera already installed in every classroom, the only hardware to purchase is a presenter remote and a USB drive.

Item	Cost
Wireless presenter remote	~$20
USB flash drive 32GB	~$10
Total one-time hardware	**~$30**

6.2 API cost per lecture (Version 3.0)

The figures below are for a typical lecture with 30 photos, 4 notes (N/O keys), and 2 screenshots (S key), using the hybrid engine. Camera captures use Gemini Flash-Lite; everything else uses Claude Sonnet or Haiku.

API Call	Model	Notes	Cost
Session naming	Haiku	Tiny prompt	~$0.00
30 camera captures — read pass	Gemini	Full frame + content-aware crops	~$0.014
30 camera captures — solve pass	Gemini	Builds structured notes	~$0.021
Smart Lookback (gated)	Sonnet	Runs on ~8 of 30 captures	~$0.204
4 text notes (N key)	Sonnet	—	~$0.040
4 photo notes (O key)	Sonnet	Includes board photo	~$0.058
2 screenshots (S key)	Sonnet	Read + solve + verification pass	~$0.179
Final lecture summary	Sonnet	—	~$0.165
Folder rename	Haiku	Tiny prompt	~$0.00
HTML generation (3 files)	—	No API call	FREE
Total per lecture (Version 3.0)			**~$0.68**

> For comparison, the same lecture cost roughly **$1.70** in Version 2.0. The hybrid engine and gated Smart Lookback together cut the typical per-lecture cost by about **60%**.

6.3 What $20 of API credit gets you

Photos per Lecture	Notes + Screenshots	Cost per Lecture	Lectures from $20	Weeks (2x/week)
30 photos	4 notes, 2 screenshots	~$0.68	~29 lectures	~14 weeks
30 photos	0 notes, 0 screenshots	~$0.20	~99 lectures	~49 weeks
50 photos	4 notes, 2 screenshots	~$0.83	~24 lectures	~12 weeks

> **Tip:** the biggest single cost driver is Smart Lookback. If you are watching your budget, set `SMART_LOOKBACK_REPAIR = False` for routine lectures and re-enable it for important ones where note quality matters most. Keeping a Gemini key in `config.txt` is the other large saving — without it, camera captures fall back to Claude Sonnet and cost roughly five times as much.

7. Recovery — If the Laptop Closes Early

7.1 How data is protected

LectureLens saves backup notes to the USB drive after every single AI response. If the laptop is closed, the power goes out, or the program crashes, all completed notes are already on the USB as text files in the `backups/` folder. No internet connection is required to recover them.

Data type	Saved?	When
Raw photos (`images/`)	Yes — always	Instantly when SPACE is pressed
Screenshots (`images/`)	Yes — always	Instantly when S is pressed
AI notes (`backups/`)	Yes — always	As soon as the AI responds (~5 sec)
Instructor notes (`backups/`)	Yes — always	As soon as the note is saved
Smart Lookback results (`backups/`)	Yes — always	After each lookback check
3 HTML files	Only if ESC was pressed	Generated at end of session
Photos still in the queue	No	Lost if the laptop closes mid-processing

7.2 Using Recover.exe

1. Plug the USB into any Windows computer
2. Double-click `Recover.exe`
3. A list of all sessions appears — pick the one to recover
4. Recover.exe reads all backup text files and builds 3 HTML files automatically
5. No internet connection or API key is required
6. HTML files appear in the session folder, ready to open in any browser

> **Improved in Version 3.0.** The recovery tool now correctly rebuilds notes that were cleaned by Smart Lookback — earlier versions could prepend stray metadata lines to those notes. It also handles every backup file format robustly, including older or partially-written files, and no longer mistakes dollar amounts in a note (such as "$5") for math.

> The summary HTML created by Recover.exe is a compiled version of all backup notes rather than a freshly AI-generated summary. The content is the same — it is simply assembled without a new Claude call.

8. Cameras & Clickers

8.1 Cameras

LectureLens automatically scans for connected USB cameras when it starts. In the normal classroom setup there is just one — the room's document camera — and the program uses it without any action needed. The teacher simply aims it at the whiteboard, or places a worksheet or textbook page under it.

If a second USB camera ever happens to be connected, both are listed in the terminal and pressing **TAB** toggles between them; the live preview shows which camera is active, and captures always use the active one. The R and T rotation keys are useful when capturing a page that is not aligned with the camera (see section 5.5).

8.2 Multiple clickers & student participation

Any number of USB presenter remotes or iClicker devices can be plugged in simultaneously. Windows merges all USB keyboard input, so the program sees every press regardless of which device sent it. A 3-second cooldown prevents duplicate captures if two people press at the same time.

Device	Trigger button
Logitech wireless presenter remotes	Right arrow button
iClicker remotes	F5 button
Any USB keyboard remote	Spacebar

A common setup: the teacher holds one clicker for key moments, while one or two students hold additional clickers to capture moments they want noted. All captures from all clickers are identical and appear in the same HTML files.

9. Troubleshooting

Problem	Likely Cause	Solution
Exe does not open	Program still running in the background	Close the terminal window first, then try again
No camera detected	Doc camera off, unplugged, or in use by another app	Switch the doc camera on, check its USB connection, close other camera apps, then restart the exe
"Gemini key not found" at startup	No Gemini key in config.txt	Optional — captures still work via Claude. Add `GEMINI_API_KEY=` to config.txt to lower cost
Missing Anthropic API key error	config.txt missing or key not set	Add `ANTHROPIC_API_KEY=your_key` to config.txt on the USB drive
No HTML files generated	Session ended before the AI finished	Use Recover.exe to rebuild all 3 HTML files from backups
Smart Lookback removing too many captures	Threshold too aggressive	Set `SMART_DELETE_WORSE_CAPTURES = False` to keep all captures
Captures look rotated or sideways	Camera mounted at an angle	Press R or T to rotate future captures into the correct orientation
N/O/S key does nothing	Preview window focused instead of the terminal	Click the terminal window once, then press the key
Note auto-cancelled immediately	Only spaces typed — spaces do not reset the timer	Type at least one real letter or number to activate the timer
Math not rendering in HTML	JavaScript disabled in the browser	Enable JavaScript, or open the file in Chrome or Edge
Preview window is black	Camera in use by another app	Close any other camera apps, then restart the exe

10. Quick Start Guide for Teachers

This page can be printed and kept with the USB drive for reference.

Starting a session

- Plug in the USB drive and presenter remote; confirm the classroom doc camera is on
- Double-click `LectureLens.exe` on the USB drive
- Wait for `Camera ready. Waiting for input...` to appear
- Check the live preview window to confirm the board is visible

During the lecture

- **SPACE** or remote clicker = capture what is on the board
- **S** = screenshot the full screen (slides, websites, documents)
- **N** = type a text question — Claude answers without seeing the board
- **O** = type a question — Claude sees the board and answers with context
- **R / T** = rotate future captures if the camera or a page is not aligned
- **TAB** = switch cameras if a second USB camera is connected
- The preview shows `Done: X Queue: Y` so you always know the status

Ending a session

- Press **ESC** — the program waits for all captures to finish processing
- Wait for `All captures processed!` and the HTML generation messages
- Press Enter when prompted — the program closes
- Open the `sessions/` folder on the USB to find your 3 HTML files
- Open any `.html` file in Chrome or Edge for full math rendering
- To print: Ctrl+P, then Save as PDF in the browser
- Unplug the USB drive safely

> **If the laptop closes early:** plug the USB into any computer and double-click `Recover.exe` to rebuild all 3 HTML files from the saved backup notes. No internet required.